Clustering travel diaries in Athens, Greece: Application of a compound-state optimal matching strategy

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Abstract Activity-based approaches to travel demand modeling have shifted focus from trips to the sequences of activities individuals pursue throughout the day. Optimal matching provides a flexible framework for quantifying the dissimilarity between such sequences, but its application to multidimensional data, where each activity is defined by multiple interrelated attributes, such as its purpose, the associated transport mode, and time of day, remains challenging. Existing approaches either decompose the problem by assuming cross-domain independence, thereby ignoring attribute interactions, or employ specialized heuristics that depend on manual parameters, do not accommodate activity-dependent costs, or compute them separately in each domain. Although direct alignment of compound sequences remains simple and exact, it remains underexplored, in part due to tractability concerns regarding the dimensionality expansion caused by the compounding process. This paper demonstrates that direct alignment is viable when this expansion is controlled through context-aware activity purpose and transport mode aggregation, combined with demand-aware temporal stratification. Furthermore, we derive substitution costs based on empirical transition rates between time-specific activity-trip combinations, thereby capturing cross-domain interactions. Applying our strategy to 512 daily travel diaries from Athens, Greece, we obtain a ten-cluster hierarchy and qualitatively examine the results using a purpose-built visualization. The solution differentiates behaviorally coherent groups, including daytime students and workers, individuals with evening schedules, and respondents with mixed activity patterns. In addition, this distinction is based on both mode preference and timing, providing preliminary evidence that our approach can capture behavioral variation in practical applications effectively.

Keywords: Activity-based modeling · Disaggregate demand · Travel diaries · Sequence analysis · Optimal matching · Hierarchical clustering.

1 Introduction

Efficient transport systems rely on planning capable of predicting and responding to changes in demand. In travel demand modeling, the emergence of activity-based approaches has represented a significant shift from traditional trip-based methods [5], such as the typical four-stage model [10]. This paradigm considers travel to be derived demand from the activities individuals pursue throughout the day [5], subject to societal [6] and spatiotemporal constraints [14,8]. Whereas trip-based models treat individual tours as independent units, activity-based approaches aim to capture behavioral interdependencies and temporal structure that link them into coherent patterns. This perspective enables improved understanding of how individuals organize their activities and respond to changes in the transport system.

The classical approach to adopting this paradigm with standard panel data, such as travel diaries, is to transform them into time-discrete sequences [18]. In transport, such a sequence, also called an activity-trip sequence, comprises a number of time-ordered tokens, each representing the behavioral state of an individual during a certain period. Although each state therefore encodes an activity or a trip, or even a location [24], it may generally consist of various attributes that collectively describe the corresponding action [9,32]. In social science, one approach to comparing these sequences is through sequence analysis, which identifies (dis-)similarities between sequences [11] as indicators of their behavioral proximity. This method was originally described in detail in [18], later adapted to social science [2], and ultimately to transport [31]. A popular analysis technique in transport literature is optimal matching [7,22,27], which quantifies dissimilarities [29].

In particular, optimal matching computes the minimum cost of transforming one sequence into the other through a series of three elementary operations: state substitution, insertion, and deletion. Because they are symmetric, insertions and deletions are often collectively referred to as indels [20]. Each operation incurs a penalty that reflects the degree of dissimilarity that operation introduces. Substitution costs quantify the difference between two states at the same position along the compared sequences, whereas indels penalize temporal state shifts [21]. These costs can be state-independent, motivated by domain knowledge or relevant assumptions, or derived from observed transition rates [29]. The total transformation cost is the ultimate measure of dissimilarity between the sequences [18]. This quantity is commonly referred to as a "distance", but whether it is indeed a metric depends on the cost definition [18].

Crucially, cost specification is a critical methodological aspect that can significantly affect results in downstream applications [1]. The simplest approach is to assign uniform penalties [18], ignoring state context and interdependencies. Alternatively, costs can be derived from observed transition rates under the assumption that states which frequently transition into one another are behaviorally proximate [20]. Although data-driven, this method commonly disregards temporal context because it uses aggregate frequencies across entire sequences, conflating identical transitions occurring at different times of day [20]. Dynamic Hamming matching [21] addresses this concern through the use of position-specific costs, which are inherently time-sensitive [28], but prohibits indels due to their similar time-warping behavior [16]. Consequently, this approach is intolerant of temporally offset but otherwise identical structures. A further challenge arises from the multidimensional nature of travel behavior, where states are compound objects consisting of multiple interdependent attributes. Standard multidomain sequence analysis provides three methods for this purpose [23]. The first approach is direct compound sequence matching, which captures cross-domain associations but may suffer from sparsity issues when data-driven costs are used, as all state components are concatenated into a single space. Therefore, the latter two methods handle this problem by either specifying compound costs as conical combinations of domain-specific penalties or matching each domain separately before aggregating the resulting dissimilarities. However, both approaches assume inter-domain independence at the state and sequence level, respectively. In transport modeling, these independence assumptions do not hold since the relevant domains are interdependent [15]. Although several field-specific methods address multidimensionality more efficiently, they remain heuristic or dependent on manual parameters [17,19], do not accommodate state-dependent costs [15], or do not allow them to interact across domains [33]. Direct compound sequence matching avoids these limitations and is both simple and exact, but remains underexplored, perhaps due to sparsity concerns [27] or its inability to partially match states across domains [17].

This paper demonstrates that this approach is indeed viable when sparsity is controlled through state aggregation and temporal stratification. In particular, we integrate activity purpose, transport mode, and time into a unidimensional compound-state optimal matching strategy that captures inter-domain interactions through transition-based costs, while remaining time-sensitive. Moreover, we avoid the limitations of dynamic Hamming matching by accommodating minor timing variations through indels. We evaluate our method by computing pairwise dissimilarities for 512 daily travel diaries collected in Athens, Greece, and clustering them into behaviorally distinct groups. The remainder of this paper is organized as follows. Section 2 describes the study dataset, formally introduces the proposed strategy, and presents the adopted clustering approach. Section 3 examines and interprets the resulting clusters using a purpose-built visualization. Finally, Section 4 discusses the findings and limitations of our method, and outlines potential directions for future research.

2 Data and Methods

2.1 Dataset Description

Our dataset derives from a small-scale travel survey of the Athens metropolitan area conducted online in 2022 as part of an undergraduate thesis. The survey was promoted via the internet and radio by the Hellenic Broadcasting Corporation and collected 513 travel diaries. Despite its size, this survey has proven useful for individual trip clustering [3]. Each diary contains six demographic attributes: sex, age, education level, employment status, monthly income, and car ownership status (Figures 1 and 2). Due to space constraints, Figure 2 shows only the four pairs with the highest corresponding bias-corrected Cramér's V [4]. For this purpose, age was discretized into 10-year bins, as per the 2021 Greek Population and Housing Census [13]. Because the survey was anonymous, respondents were not required to provide this information. Consequently, only 461 out of the original 513 diaries include complete demographics.

Each diary also records the respondent's home zone, selected from a partition of the study area into 35 regions plus an additional virtual zone representing exterior locations. Furthermore, each diary contains up to five trips the respondent made on a typical workday, including a final return-home trip. Each trip is defined by its destination zone, the purpose of the destination activity, the travel mode, and the departure time. The available purposes are home, education, work, market, recreation, service, and other. The available modes are car, motorcycle, taxi, bus, train, bicycle, e-scooter, and walk. Due to privacy constraints, departure times were reported as one of seven intervals: six consecutive three-hour windows spanning 05:00–23:00, and a single six-hour overnight window between 23:00 and 05:00.

To construct coherent trip chains, we assumed that each diary's first trip originates from the respondent's home zone and that each subsequent trip originates from the preceding destination. To obtain concrete departure times for sequence construction, we uniformly sampled a time within the reported interval, subject to trip-chain feasibility constraints and a minimum activity duration of 30 minutes. The resulting departure times are therefore synthetic rather than directly observed. Travel times between zone centroids were obtained from the Google Routes API for car and public transit. These data were collected on Thursday, 22 May 2025, between 00:00 and 24:00, with a two-hour sampling frequency. Travel times for all other hours were linearly interpolated. Travel times for other modes were estimated by scaling car travel times by average speed ratios for the study area. During the scheduling process, only one diary was deemed completely infeasible and consequently discarded.

In addition, we imputed missing return-home trips in diaries where the final reported purpose was neither home nor recreation. Therefore, recreation activities were permitted to extend beyond the monitoring period. For imputation, we used the last reported mode and generated a valid activity duration via inverse transform sampling. Finally, we cropped all diaries exceeding 24 hours to the observation window. The resulting sample comprises 512 diaries.

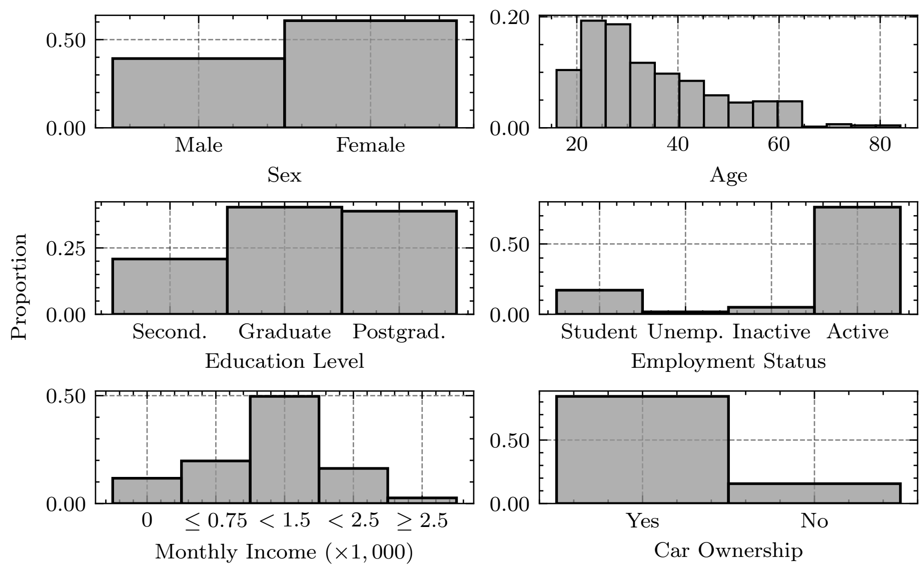

Figure 1. Marginal distributions of the demographic attributes comprising our dataset.

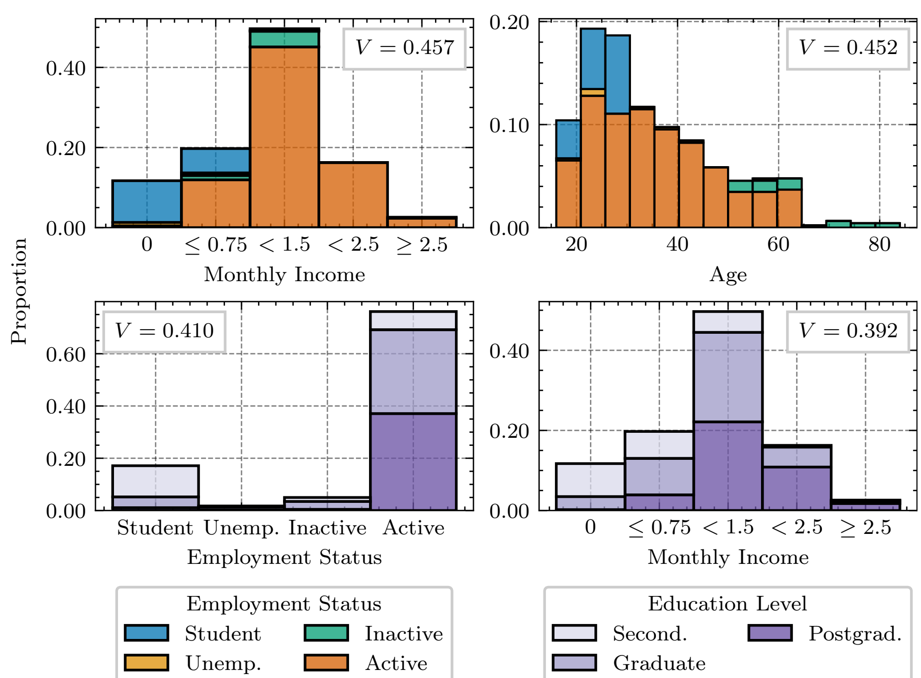

Figure 2. Bivariate distributions of selected demographic attributes comprising our dataset.

2.2 Sequence Representation

For optimal matching, we transform each diary from its original representation into a discrete state sequence. This transformation comprises three stages: constructing a time-continuous episodic representation, discretizing time into fixed-length intervals, and augmenting the corresponding states with temporal information.

Let $T \in \mathbb{R}_{>0}$ denote the duration of the observation period in hours. For our dataset, $T = 24$ hours. To capture overnight activities, we anchor the window at a clock time of 04:00. Formally, the diary time $\tau \in [0, T]$ corresponds to clock time $(\tau + 4) \mod 24$. Thus, a diary time of 00:00 i.e., its start time, corresponds to 04:00, and 24:00 i.e., its end time, to 04:00 on the following day.

A diary records a sequence of episodes, where each episode represents a period during which the respondent continuously engages in a particular activity or trip. Let $\mathcal{A} = \mathcal{P} \cup \mathcal{M}$ denote the alphabet of possible states, defined as the union of activity purposes $\mathcal{P}$ and travel modes $\mathcal{M}$. These sets are disjoint by construction. In practice, the examined diaries contain seven purposes and eight modes, yielding a total of 15 distinct states. Therefore, an episode is a triplet (s, α, β), consisting of a state $s \in \mathcal{A}$, a start time $\alpha \in [0, T)$, and an end time $\beta \in (0, T]$ with $\alpha < \beta$. Then, a diary $D = \langle (s_1, \alpha_1, \beta_1), \ldots, (s_m, \alpha_m, \beta_m) \rangle$ is a sequence of m such episodes ordered by start time, where $\alpha_1 = 0$, $\beta_m = T$, and $\beta_k = \alpha_{k+1}$ for all $k \in \{1, \ldots, m-1\}$. These constraints ensure that the episodes partition the observation window without gaps or overlaps.

To align with sequence analysis conventions, we next discretize the observation window into n consecutive intervals of fixed duration Δ hours. Given $\Delta > 0$, the sequence length is $n = \lceil \frac{T}{\Delta} \rceil$, where $\lceil \cdot \rceil$ denotes the ceiling function. For our dataset, $\Delta = 0.25$ hours i.e., 15 minutes, yielding $n = 96$ intervals. The i-th interval spans $[(i-1)\,\Delta, i\Delta)$ for each $i \in \{1, \ldots, n-1\}$, with the final interval $[(n-1)\,\Delta, T]$ including the endpoint. Each interval is assigned the state most prevalent within it, as measured by total overlap duration. In particular, the temporal overlap between the k-th episode and the i-th interval is:

$$\omega_k(i) = \max\left(0, \min\left(\beta_k, i\Delta\right) - \max\left(\alpha_k, (i-1)\,\Delta\right)\right) \tag{1}$$

and the state $x_i \in \mathcal{A}$ assigned to the i-th interval is:

$$x_i = \arg\max_{a \in \mathcal{A}} \sum_{k=1}^{m} \omega_k\,(i) \cdot [s_k = a] \tag{2}$$

where $[\cdot]$ denotes the Iverson bracket. Ties are resolved in favor of the earliest-starting overlapping episode. Applying this procedure to each interval yields the discrete sequence $X = \langle x_1, \ldots, x_n \rangle$.

Although sequence position implicitly encodes time, we further augment each state with explicit information about the period during which it occurs. This grouping allows us to partition the day into behaviorally meaningful intervals whose boundaries reflect typical shifts in travel demand, while limiting the resulting alphabet expansion. Specifically, we define a set of period labels $\mathcal{L} \subset \mathbb{Z}_{>0}$

and a mapping $\rho : [0, T] \to \mathcal{L}$ that assigns each diary time to its corresponding period. For our dataset, we use four periods that reflect standard peak and off-peak hours in the study area:

$$\rho(\tau) = \begin{cases} 1, & \text{if } \tau \in [0, 3) \cup [15, 24] \\ 2, & \text{if } \tau \in [3, 6) \\ 3, & \text{if } \tau \in [6, 12) \\ 4, & \text{if } \tau \in [12, 15) \end{cases} \tag{3}$$

These intervals correspond to nighttime off-peak hours at 04:00–07:00 and 19:00–04:00, an AM peak at 07:00–10:00, the corresponding off-peak at 10:00–16:00, and a PM peak at 16:00–19:00. The period assigned to the i-th interval is determined by its start time: $p_i = \rho\left((i-1)\,\Delta\right)$.

This grouping yields the augmented alphabet $\tilde{\mathcal{A}} = \mathcal{A} \times \mathcal{L}$, with cardinality $|\tilde{\mathcal{A}}| = 15 \cdot 4 = 60$. The augmented sequence is $\tilde{X} = \langle (x_1, p_1), \ldots, (x_n, p_n) \rangle$, where each element is a state-period pair that jointly encodes which activity or trip is currently occurring and the corresponding period.

2.3 Dissimilarity Computation

To mitigate sparsity in transition probability estimation, we aggregate behaviorally similar states prior to deriving substitution and indel costs. In particular, activity purposes are consolidated into three categories: home, rigid, and flexible, where the second group comprises education and work, and the third all other out-of-home purposes. This distinction was inspired by [9,27] and reflects the assumption that the timing of rigid activities is constrained by various external factors, whereas flexible activities can generally be scheduled at the individual's discretion. In this context, home activities are thought to serve as spatial anchors with respect to which other activities are organized. Furthermore, taxi is merged with car, and bicycle and e-scooter are combined into a single micromobility category. This approach reflects typical behavioral patterns in the study area, where the aggregated modes are commonly used interchangeably. Ultimately, this grouping yields the reduced alphabet $\tilde{\mathcal{A}}'$ with cardinality $|\tilde{\mathcal{A}}'| = (3 + 6) \cdot 4 = 36$. All subsequent calculations operate on this alphabet.

Hypothesizing that states with high mutual transition rates are behaviorally proximate, we derive substitution costs from observed state dynamics. Formally, the transition rate from a state $s \in \tilde{\mathcal{A}}'$ to a distinct state $s' \in \tilde{\mathcal{A}}', s' \neq s$ is the empirical conditional probability:

$$P\left(s' \mid s\right) = \frac{\text{TransitionCount}\left(s, s'\right)}{\sum_{s'' \neq s} \text{TransitionCount}\left(s, s''\right)} \tag{4}$$

where $\text{TransitionCount}\colon \tilde{\mathcal{A}}' \times \tilde{\mathcal{A}}' \to \mathbb{R}_{\geq 0}$ counts the transitions from one state to another across all sequences. In contrast to the standard formulation [12,27], we do not count self-transitions to separate state persistence, which reflects episode duration in discretized sequences, from transition behavior. Because either sequence could serve as the source of transformation, a well-defined dissimilarity measure requires symmetric costs. Since transition rates are inherently asymmetric, we define the substitution cost as:

$$c\left(s, s'\right) = \begin{cases} 0, & \text{if } s = s' \\ 2 - P\left(s' \mid s\right) - P\left(s \mid s'\right), & \text{otherwise} \end{cases} \tag{5}$$

This formulation is symmetric by construction and bounded in $[0, 2]$. We acknowledge that high transition rates may reflect complementarity rather than substitutability. For instance, home and work activities frequently follow one another despite being behaviorally distinct. However, this study prioritizes empirical regularity over theoretical correctness.

In contrast to the substitution cost, we define the indel cost as a state-independent constant to control the trade-off between timing fidelity and pattern similarity. To illustrate this trade-off, consider that indels alter sequence length in opposite directions. Therefore, any transformation between equal-length sequences must include these operations in equal numbers. Consequently, indels allow states at different positions to be compared. For instance, transforming $\langle 1, 2, 3, 4\rangle$ into $\langle 1, 3, 4, 5\rangle$ via substitution alone requires three operations. However, deleting 2 from and appending 5 to the former sequence achieves the same result in two operations by matching 3 and 4 across misaligned positions. This flexibility causes sequences with similar activity patterns but slightly offset timing to appear more alike. In this context, we define the indel cost as:

$$\gamma = \frac{1}{2} \max\left(c\right) = 1 \tag{6}$$

ensuring that substitution is never more costly than the corresponding indel alternative and that direct state comparisons are preferred when states are temporally aligned. Since the period boundaries defined in Equation 3 are themselves approximations and transitions between them are temporally fuzzy, permitting limited temporal flexibility also accommodates timing variations near them.

Finally, we perform optimal matching using a generalized Wagner-Fischer algorithm [30] that accommodates state-dependent substitution costs. Because the algorithm operates on sequences of atomic symbols, we encode each augmented state $(x_i, p_i) \in \tilde{X}$ as a compound label by concatenating its components. The results are compiled into a symmetric dissimilarity matrix.

2.4 Hierarchical Clustering

To segment the sequences into behaviorally distinct groups, we apply agglomerative clustering to the dissimilarity matrix. We specifically selected this approach for two reasons. First, it does not assume that the input matrix is a valid metric. Although relevant properties are enforced by construction, we do not claim that our cost definition produces a valid distance. Second, hierarchical clustering captures the nested structure of travel demand, whereby broad archetypes are divided into specific subgroups based on timing and mode choice.

Hierarchical clustering requires a linkage criterion that determines how inter-cluster dissimilarity is quantified. Since we do not assume the metric property, we only consider methods that do not require Euclidean distances. Ultimately, we select average linkage [25] because it maximizes the cophenetic correlation coefficient [26] of the resulting dendrogram. In this way, we ensure that our solution captures pairwise dissimilarities as accurately as possible.

Finally, we visualize the solution using a truncated dendrogram, vertically arranged by merge order. Each node of the dendrogram is represented by the temporal state distribution of the corresponding cluster. This representation, adapted from [12,28], enables simultaneous inspection of cluster composition and hierarchy. The construction and interpretation of this visualization are detailed in Section 3.

3 Results

The clustering solution is presented as a dendrogram in Figure 3. To aid with visualization, the dendrogram has been truncated to show the last ten clusters formed as leaf nodes, with all earlier merges collapsed into them. The choice of ten clusters was informed by prior knowledge of typical travel behavior in the study area [3].

Each node is depicted by the temporal state distribution of the corresponding cluster. The top panel shows travel modes, whereas the bottom panel shows activity purposes. In addition, cluster sizes and a numbering of the leaves are provided. Finally, each node is annotated with the corresponding linkage distance $\tilde{H}$, scaled to equal one at the root. This quantity represents the dissimilarity at which the corresponding cluster was formed. The nodes are vertically arranged by merge order such that any solution containing fewer clusters than the total number of leaves can be obtained by successively discarding leaves from the bottom of the dendrogram. For instance, the nine-cluster solution comprises Clusters 1–8 and the parent of Clusters 9 and 10. This approach facilitates simultaneous inspection of all “coarser” solutions.

The root state distribution indicates that respondents collectively tend to start the day at home, depart for education or work approximately between 08:00 and 11:00, and remain there until approximately 17:00–21:00 before returning home or engaging in other activities. The width of these intervals implies relatively high variability in work schedules across our dataset. For commuting, car and public transport, especially train, are the primary travel modes. In contrast, flexibly timed activities are strongly associated with walking, as evidenced by its increased share between 18:00 and 00:00. The overall travel demand appears to peak during commute hours, although secondary peaks appear in the early morning, mid-afternoon, and mid-evening.

Concerning individual partitions, Cluster 9 is the largest, representing more than 40% of all respondents. This cluster displays a standard daytime employ-

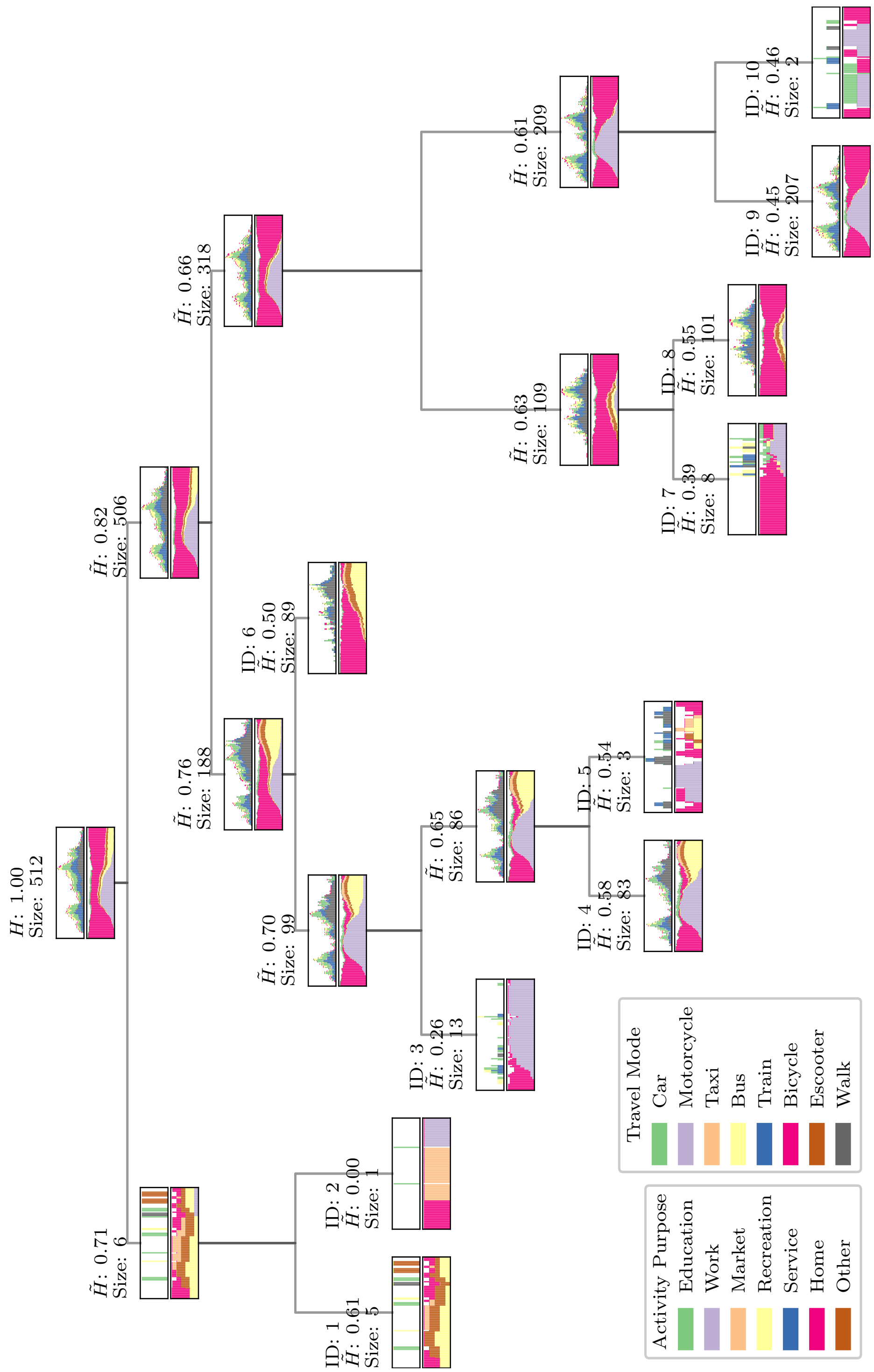

Figure 3. Truncated dendrogram with embedded temporal state distributions.

ment pattern, with work being the dominant activity from around 08:00 to 20:00. Full-time students are also represented in this cluster, albeit in significantly lower numbers, requiring a “finer” solution to differentiate. Following global patterns, mode share is dominated by car and public transport, with train being significantly more popular than bus, although relatively few respondents choose to commute on foot.

Cluster 4 also captures students and workers, comprising approximately 16% of our dataset. However, these respondents engage in evening after-work activities, particularly recreation, for significantly longer periods that extend well beyond midnight. Most members walk to the locations of these activities before returning home by car, as evidenced by the corresponding evening peak.

In addition, Clusters 6 and 8 together represent approximately 37% of respondents and illustrate how our compound-state approach captures behavioral heterogeneity beyond timing. In particular, Cluster 8 represents mode preferences that differ from the car-dominant behavior exhibited in Cluster 9. Although this cluster also includes workers and students with standard daytime schedules, they rely more on public transit and walking than car for commuting. This cluster also captures respondents with flexible schedules who engage in other out-of-home activities, such as running errands and shopping, but otherwise remain at home. Cluster 6 captures a similar pattern but shifted to evening and night hours.

Moreover, Clusters 3 and 7 exhibit irregular education and work patterns. On the one hand, the former group consists of respondents with extended working hours or relatively long-distance commutes. Yet, the associated morning peak remains unchanged, suggesting extended work durations rather than shifted schedules. On the other hand, the latter cluster primarily represents evening students and night shift workers, whose travel exhibits notable modal diversity, with bus, car, train, and walking all represented.

Finally, several clusters are particularly small, indicating outlier or residual patterns. Most notably, Cluster 2 is a singleton exhibiting almost continuous shopping activity throughout the standard store hours in Greece, followed by overnight work. Since most night-shift workers (Clusters 3 and 7) typically remain at home during daytime, this pattern is isolated. Clusters 5 and 10 both display what appear to be part-time work or education patterns, either in the morning or the evening. Cluster 1 contains sequences with potential data quality issues, since it captures patterns that are either constant or oscillate between home and a limited number of unusually timed activities throughout the observation period.

4 Conclusion

This paper proposes and demonstrates a compound-state optimal matching strategy for daily travel diary clustering. By integrating activity purpose, travel mode, and time into a unified framework with transition-based costs, we capture cross-domain interactions exactly and simply. These interactions would otherwise be lost or require complex methods with their own drawbacks. Our experimental results provide preliminary evidence that our approach can successfully distinguish behavioral patterns based on both timing and mode share. However, quantitative analysis and formal validation remain priorities for future work.

In addition, our visualization of the clustering solution supports several downstream applications. For population segmentation, leaf clusters can be cross-tabulated with relevant covariates (e.g., employment status, schedule constraints, and mode access) to further investigate travel demand in the study area. For outlier detection and data quality control, relatively small and singleton clusters can be easily identified and either discarded or investigated further. For sensitivity analysis, coarser solutions can be directly compared. In this context, we note that in agglomerative clustering, the most important splits, as identified by the particular linkage criterion, generally appear higher in the corresponding dendrogram [28]. We also emphasize the inherent ability to study any visible cluster and obtain solutions that correspond to sophisticated dendrogram cuts that could otherwise be complicated to obtain. For instance, depending on the particular application, any two clusters may be selectively and iteratively merged to the required resolution, or whole subtrees may be discarded. Beyond methodological applications, the clusters themselves provide key messages and actionable evidence for efficient transport planning, supporting targeted policy interventions for specific population segments.

Although our approach and subsequent analysis show promise, we acknowledge several limitations that suggest directions for future research. First, the statistical representativeness of our dataset remains unverified, either in terms of population coverage or sample size. Replication with larger samples in other regions would assess generalizability. Second, the survey limited the number of reported trips, potentially failing to capture complex patterns. Future surveys should relax this constraint. Third, trip departure times are synthetic, derived through constrained uniform sampling of predefined intervals. This introduces uncertainty near peak transitions and fails to capture behavioral idiosyncrasies. Collecting precise departure times would improve accuracy. Fourth, transition-based substitution costs assume that co-occurring states are co-substitutable. This assumption does not hold for complementary pairs such as home and work. More sophisticated cost specifications could address this limitation. Moreover, although consistent with relevant prior work [3], our clustering solution has not been validated against external factors. Cross-tabulation with demographic attributes would enable direct application in targeted planning and policy-making. Additionally, a comprehensive sensitivity analysis comparing our temporal stratification and cost derivation schemes with alternatives e.g., [27] would strengthen the methodological contribution. Finally, methodological extensions such as including more dimensions e.g., activity duration, geographical location, and trip distance, in state definitions could be explored.

Acknowledgments. ***

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Abbot, A., Tsay, A.: Sequence analysis and optimal matching methods in sociology: Review and prospect. Sociological Methods & Research **29**(1), 3–33 (2000). https://doi.org/10.1177/0049124100029001001, https://doi.org/10.1177/0049124100029001001
2. Abbott, A., Forrest, J.: Optimal matching methods for historical sequences. The Journal of Interdisciplinary History **16**(3), 471–494 (1986), http://www.jstor.org/stable/204500
3. Andrinopoulou, E., Tzouras, P.G.: Applying spectral clustering to decode mobility patterns in athens, greece. Applied Sciences **15**(7) (2025). https://doi.org/10.3390/app15073419, https://www.mdpi.com/2076-3417/15/7/3419
4. Bergsma, W.: A bias-correction for Cramér's V and Tschuprow's T. Journal of the Korean Statistical Society **42**(3), 323–328 (2013). https://doi.org/https://doi.org/10.1016/j.jkss.2012.10.002, https://www.sciencedirect.com/science/article/pii/S1226319212001032
5. Bhat, C.R., Koppelman, F.S.: Activity-Based Modeling of Travel Demand, pp. 35–61. Springer US, Boston, MA (1999). https://doi.org/10.1007/978-1-4615-5203-1_3, https://doi.org/10.1007/978-1-4615-5203-1_3
6. Chapin, F.S.: Free time activities and quality of urban life. Journal of the American Institute of Planners **37**(6), 411–417 (1971). https://doi.org/10.1080/01944367108977391, https://doi.org/10.1080/01944367108977391
7. Crawford, F., Watling, D., Connors, R.: Identifying road user classes based on repeated trip behaviour using bluetooth data. Transportation Research Part A: Policy and Practice **113**, 55–74 (2018). https://doi.org/https://doi.org/10.1016/j.tra.2018.03.027, https://www.sciencedirect.com/science/article/pii/S096585641730352X
8. Cullen, I., Godson, V.: Urban networks: The structure of activity patterns. Progress in Planning **4**, 1–96 (1975). https://doi.org/https://doi.org/10.1016/0305-9006(75)90006-9, https://www.sciencedirect.com/science/article/pii/0305900675900069
9. Dharmowijoyo, D.B., Susilo, Y.O., Karlström, A.: Analysing the complexity of day-to-day individual activity-travel patterns using a multidimensional sequence alignment model: A case study in the bandung metropolitan area, indonesia. Journal of Transport Geography **64**, 1–12 (2017). https://doi.org/https://doi.org/10.1016/j.jtrangeo.2017.08.001, https://www.sciencedirect.com/science/article/pii/S096669231630045X
10. de Dios Ortúzar, J., Willumsen, L.G.: Modelling Transport. John Wiley & Sons, Ltd (2011)
11. Elzinga, C.H.: Sequence analysis: Metric representations of categorical time series. techreport, Vrije Universiteit Amsterdam (2008)
12. Gabadinho, A., Ritschard, G., Müller, N.S., Studer, M.: Analyzing and visualizing state sequences in r with traminer. Journal of Statistical Software **40**(4), 1–37 (2011). https://doi.org/10.18637/jss.v040.i04, https://www.jstatsoft.org/index.php/jss/article/view/v040i04

13. Hellenic Statistical Authority: 2021 Population and Housing Census: Resident population by age groups, citizenship, place of birth, legal marital status and size and composition of households. e-Publication (Aug 2023), https://elstat-outsourcers.statistics.gr/Booklet_ÎŚÏĂÎ¿ÏĎÎµÎżÎµÏČÎijÎśÏĎÎśÎăÎżÎůÎÿÏĚÏČÎijÎ¿ÏĚ2023_IIEN_FINAL2_WEB.pdf
14. Hägerstrand, T.: What about people in regional science? Papers of the Regional Science Association **24**(1), 6–21 (1970). https://doi.org/10.1007/BF01936872, https://doi.org/10.1007/BF01936872
15. Joh, C.H., Arentze, T., Hofman, F., Timmermans, H.: Activity pattern similarity: a multidimensional sequence alignment method. Transportation Research Part B: Methodological **36**(5), 385–403 (2002). https://doi.org/https://doi.org/10.1016/S0191-2615(01)00009-1, https://www.sciencedirect.com/science/article/pii/S0191261501000091
16. Joh, C.H., Arentze, T.A., Timmermans, H.J.P.: A position-sensitive sequence-alignment method illustrated for space–time activity-diary data. Environment and Planning A: Economy and Space **33**(2), 313–338 (2001). https://doi.org/10.1068/a3323, https://doi.org/10.1068/a3323
17. Joh, C.H., Arentze, T.A., Timmermans, H.J.: Multidimensional sequence alignment methods for activity-travel pattern analysis: A comparison of dynamic programming and genetic algorithms. Geographical Analysis **33**(3), 247–270 (2001). https://doi.org/https://doi.org/10.1111/j.1538-4632.2001.tb00447.x, https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1538-4632.2001.tb00447.x
18. Kruskal, J.B.: An overview of sequence comparison: Time warps, string edits, and macromolecules. SIAM Review **25**(2), 201–237 (1983), http://www.jstor.org/stable/2030214
19. Kwan, M.P., Xiao, N., Ding, G.: Assessing activity pattern similarity with multidimensional sequence alignment based on a multiobjective optimization evolutionary algorithm. Geographical Analysis **46**(3), 297–320 (2014). https://doi.org/https://doi.org/10.1111/gean.12040, https://onlinelibrary.wiley.com/doi/abs/10.1111/gean.12040
20. Lesnard, L.: Optimal Matching and Social Sciences (Jan 2006), https://shs.hal.science/halshs-00008122, working paper or preprint
21. Lesnard, L.: Setting cost in optimal matching to uncover contemporaneous socio-temporal patterns. Sociological Methods & Research **38**(3), 389–419 (2010). https://doi.org/10.1177/0049124110362526, https://doi.org/10.1177/0049124110362526
22. Liu, F., Gao, Z., Janssens, D., Jia, B., Wets, G., Yang, Y.: Identifying business activity-travel patterns based on gps data. Transportation Research Part C: Emerging Technologies **128**, 103136 (2021). https://doi.org/https://doi.org/10.1016/j.trc.2021.103136, https://www.sciencedirect.com/science/article/pii/S0968090X21001558
23. Ritschard, G., Liao, T.F., Struffolino, E.: Strategies for multidomain sequence analysis in social research. Sociological Methodology **53**(2), 288–322 (2023). https://doi.org/10.1177/00811750231163833, https://doi.org/10.1177/00811750231163833
24. Shoval, N., McKercher, B., Birenboim, A., Ng, E.: The application of a sequence alignment method to the creation of typologies of tourist activity in time and space. Environment and Planning B: Planning and Design **42**(1), 76–94 (2015). https://doi.org/10.1068/b38065, https://doi.org/10.1068/b38065
25. Sokal, R.R., Michener, C.D.: A statistical method for evaluating systematic relationships. University of Kansas science bulletin **38**, 1409–1438 (1958), https://api.semanticscholar.org/CorpusID:61950873

26. Sokal, R.R., Rohlf, F.J.: The comparison of dendrograms by objective methods. Taxon **11**(2), 33–40 (1962), http://www.jstor.org/stable/1217208
27. Song, Y., Ren, S., Wolfson, J., Zhang, Y., Brown, R., Fan, Y.: Visualizing, clustering, and characterizing activity-trip sequences via weighted sequence alignment and functional data analysis. Transportation Research Part C: Emerging Technologies **126**, 103007 (2021). https://doi.org/https://doi.org/10.1016/j.trc.2021.103007, https://www.sciencedirect.com/science/article/pii/S0968090X21000395
28. Studer, M.: Weightedcluster library manual: A practical guide to creating typologies of trajectories in the social sciences with r. LIVES Working Papers **024**, 1–32 (2013). https://doi.org/10.12682/lives.2296-1658.2013.24, http://dx.doi.org/10.12682/lives.2296-1658.2013.24
29. Studer, M., Ritschard, G.: What matters in differences between life trajectories: A comparative review of sequence dissimilarity measures. Journal of the Royal Statistical Society Series A: Statistics in Society **179**(2), 481–511 (07 2015). https://doi.org/10.1111/rssa.12125, https://doi.org/10.1111/rssa.12125
30. Wagner, R.A., Fischer, M.J.: The String-to-String Correction Problem. J. ACM **21**(1), 168–173 (Jan 1974). https://doi.org/10.1145/321796.321811, https://doi.org/10.1145/321796.321811
31. Wilson, W.C.: Activity pattern analysis by means of sequence-alignment methods. Environ Plan A **30**(6), 1017–1038 (Jun 1998). https://doi.org/10.1068/a301017, https://doi.org/10.1068/a301017
32. Xianyu, J., Rasouli, S., Timmermans, H.: Analysis of variability in multi-day gps imputed activity-travel diaries using multi-dimensional sequence alignment and panel effects regression models. Transportation **44**(3), 533–553 (2017). https://doi.org/10.1007/s11116-015-9666-2, https://doi.org/10.1007/s11116-015-9666-2
33. Zhai, W., Bai, X., ren Peng, Z., Gu, C.: From edit distance to augmented space-time-weighted edit distance: Detecting and clustering patterns of human activities in puget sound region. Journal of Transport Geography **78**, 41–55 (2019). https://doi.org/https://doi.org/10.1016/j.jtrangeo.2019.05.003, https://www.sciencedirect.com/science/article/pii/S0966692318305465